25901679 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCFTQ100 - BASIC BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. What are dextrins? [U] / [CO1]
2. Discuss the importance of hydrogen bonding in stabilizing α-helix. [U] / [CO2]
3. What is Ketonic rancidity? [U] / [CO4]
4. What are aldopentoses? Give examples. [K] / [CO1]
5. What are activators? Give examples. [U] / [CO3]
6. What are transfatty acids? How are they harmful to our body? [A] / [CO4]
7. Define K_m. State its significance [A] / [CO3]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8.	Prepare a note on the following Reactions: 1) Reducing property of sugars 2) Dehydration Reaction 3) Osazone Formation 4) Oxidation and Reduction	[U] / [CO1]
9.	Discuss the impact of activators in an enzyme catalysed reaction.	[E] / [CO3]
10.	Differentiate between fats and oils.	[An] / [CO4]
11.	Give the structural configuration of ketohexoses with emperical formula $C_6H_{12}O_6$.	[K] / [CO1]
12.	Describe the affect of changing pH on altering the ionic state of amino acids.	[A] / [CO2]
13.	What are Lipids? Narrate the functions of lipids.	[U] / [CO4]
14.	Explain the chemical properties of amino acids due to COOH group.	[U] / [CO2]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15.	Define Active site. Outline the major salient features of active site.	[K] / [CO3]
16.	Describe in detail on classification of carbohydrates.	[U] / [CO1]
17.	Write an essay on the structural organisation of proteins. Add a note on the chemical bonds responsible for its formation.	[U] / [CO2]
18.	Evaluate the structure, properties and functions of major phospholipids.	[A] / [CO4]

[10x2 = 20]